

ABHIJITH SIVAPRASADAN

Thermal-Fluid & Process Engineer | CFD, Heat Transfer, Thermal Energy Systems & Process Modelling

Solna, Sweden | +46 76 969 2014 | abhijithsivaprasadan@gmail.com

[linkedin.com/in/abhijith-sivaprasadan](https://www.linkedin.com/in/abhijith-sivaprasadan) | github.com/abhijith-sivaprasadan | abhijith-sivaprasadan.github.io

PROFILE

Thermal-fluid and process engineer with an M.Sc. in Sustainable Energy Engineering from KTH Royal Institute of Technology and a B.Tech in Mechanical Engineering, working across CFD and heat-transfer analysis, thermal energy systems and industrial process/energy methodology. My master thesis at Siemens Energy's Fluid Dynamics Laboratory built and verified compressible-flow and conjugate heat-transfer models of high-temperature industrial hardware in ANSYS Fluent, including a three-level mesh-independence study and an analytical heat-loss cross-check that agreed with the simulation within 0.1%. I have hands-on laboratory experience across thermal energy storage, refrigeration/HVAC and air-handling systems, and I developed a comparative industrial energy-audit methodology for Alleima AB across six countries, including KPI/EnPI baselining and an ISO 50001/EED-oriented framework. Independent cross-checking, through an analytical calculation, a second implementation, automated tests or measured data, is a recurring part of my engineering workflow.

EDUCATION

M.Sc. Sustainable Energy Engineering

Aug 2023 - 2026

KTH Royal Institute of Technology, Stockholm — 123 ECTS (120 required + 3 elective) complete per official KTH confirmation letter (24 Sep 2026); degree certificate/conferral in progress

B.Tech Mechanical Engineering

2017 - 2021

College of Engineering Perumon, APJ Abdul Kalam Technological University, India

PROFESSIONAL EXPERIENCE

Master Thesis Student, Fluid Dynamics Laboratory

Apr - Nov 2025

Siemens Energy AB, Finspång

- Built compressible CFD and conjugate heat-transfer models in ANSYS Fluent (SST k- ω) of high-temperature industrial hardware, verifying spatial-discretisation sensitivity with a three-level mesh-independence study; an analytical heat-loss consistency check agreed with Fluent within 0.1% across six CHT cases.
- Commissioned the measurement chain for the planned experimental validation campaign using NI-DAQ/NI MAX, LabVIEW and thermocouple/pressure channels; when an internal heater failure occurred during experimental preparation, before the high-temperature campaign began, conducted the root-cause analysis with the industrial supervisor and delivered a re-scoped 15-run numerical study within the original schedule.

Energy Efficiency Intern

Dec 2024 - Jun 2025

Alleima AB (remote)

- Developed a comparative industrial energy-audit methodology across six countries, including a tiered ISO 50001/EED-oriented framework, KPI/EnPI baselines and load-driver normalisation.

Engineer (backend software)

Sep 2021 - Jun 2023

QBurst Technologies, Thiruvananthapuram, India

- Developed production backend services (Go, later TypeScript/NestJS) with automated tests and negative-path validation; this reproducibility and verification discipline now underpins every engineering model and measurement comparison I run.

RELEVANT GRADUATE COURSEWORK

Numerical Heat Transfer in Energy Technology (Grade A) · Energy Storage Technology (Grade B) · Applied Heat and Power Technology · Energy in the Built Environment (IDA ICE, AHU psychrometrics, HVAC/refrigeration laboratory) · Small Scale Polygeneration · Modeling of Energy Systems – Heat and Power Generation.

THERMAL & PROCESS ENGINEERING PROJECTS

M-TES PCM discharge characterisation laboratory

KTH laboratory (MJ2386), group project

- Characterised the discharge of a 44 L/33 kg RT60 HC latent-heat storage prototype using ten PT100 sensors, a Type-T thermocouple and a Keithley DAQ over an 8-hour test; identified the storage tank's own thermal mass as the dominant unmodelled term through residual analysis against a theoretical energy balance (0.14% agreement between theoretical and measured discharge energy).

R134a refrigeration / heat-pump performance laboratory

KTH (MJ2509)

- Evaluated refrigeration and heat-pump operation at approximately -10, 0 and +10°C evaporation conditions using measured pressures, temperatures, compressor electrical input and condenser-water flow; calculated heat duties and COP and interpreted the cycle on an R134a pressure-enthalpy diagram.

Air-handling-unit psychrometrics laboratory

KTH (MJ2509)

- Evaluated cooling-coil performance at three airflow conditions using wet-/dry-bulb temperature measurements, a psychrometric chart and airflow/enthalpy calculations; identified and reported an inconsistent data point as likely measurement error rather than omitting it.

Branched hydronic distribution-network sizing

KTH (MJ2509), COMSOL

- Used COMSOL Application Builder to size and balance a branched copper-pipe hydronic distribution network, selecting pipe diameters and branch loss coefficients to achieve a target mass-flow distribution.

TES Discharge Screen: thermal-storage technology screening framework

Python, Pyomo/HiGHS, Modelica/FMI

- Built a packed-bed dynamic thermal model alongside molten-salt and PCM sub-models, cross-checked a Modelica FMU (via OpenModelica/FMPy) against an independent Python model over an 8-hour test (0.23% maximum temperature deviation); framed as a software reference study, with no experimental validation against a real plant.

OpenSteamOpt: steam-and-power optimisation and Modelica digital-twin demonstrator

Python, Pyomo/HiGHS, Modelica/FMI

- Built a two-boiler/two-header steam-and-power plant demonstrator coupling MILP dispatch optimisation with a Modelica dynamic plant model, verified via FMI 2.0 co-simulation (OpenModelica/FMPy) and cross-checked against an independent Python shadow-twin model; synthetic scenarios, not real-plant savings.

Independent software projects above (TES Discharge Screen, OpenSteamOpt) use AI-assisted implementation; I define the problem formulation, model architecture, validation criteria and interpretation of results.

TECHNICAL SKILLS

CFD and heat transfer	ANSYS Fluent (compressible flow, conjugate heat transfer, SST $k-\omega$); mesh-independence methodology; finite-volume numerical heat-transfer methods
Thermal systems	Thermal energy storage, refrigeration/HVAC cycles, psychrometrics, hydronic networks; PT100/thermocouple instrumentation, Keithley/NI-DAQ
Process & energy methodology	Industrial energy-audit methodology, KPI/EnPI baselining, ISO 50001/EED-oriented frameworks, load-driver normalisation, steam/power dispatch optimisation
CAD	ANSYS SpaceClaim/Discovery, SolidWorks; hands-on Siemens NX modelling for an intermediate legacy-component comparison
Programming	Python (NumPy, SciPy, pandas), MATLAB, Modern Fortran, Go, TypeScript; Git, automated testing

ADDITIONAL INFORMATION

Conference: Sponsored student delegate, Converging Energy Technologies (CET2026), Oskarshamn, Sweden, 28 Sep - 1 Oct 2026
 Languages: English (professional working proficiency), Malayalam (native), Swedish (actively studying)